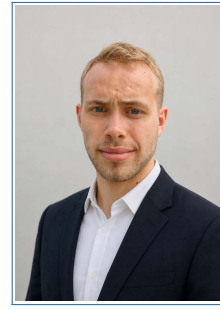


Vebjørn Erlandsen

28.05.2004

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Portfolio & Projects: <https://v3rlandsen.github.io/portfolio>



EDUCATION

M.Sc. Cybernetics and Robotics Norwegian University of Science and Technology (NTNU)

Trondheim 2026 – 2028 (ongoing) Specialization: Robots and Vessel Control

Current coursework: Linear System Theory (state-space models, LQR, Kalman filtering),
Nonlinear Control Systems, Algorithms and Data Structures

B.Sc. Automation and Electronics Design University of Stavanger 2023 – 2026

Bachelor's thesis: "Simulation-Based Testing Framework for Embedded Systems Using MQTT and Cross-Compiled Code" – Grade A

SELECTED PROJECTS

- Simulation-Based Testing Framework for Embedded Systems (Bachelor Thesis, 2026) • Designed and implemented a complete simulation framework allowing ABB's Virtual IPS controller to run closed-loop tests against a virtual process • Technologies: C++, C#, MQTT, Windows shared memory, Python
- ESP32 4-Wire Fan Controller with Live RPM & Web UI (Personal, 2026) • Closed-loop PWM speed controller with tachometer feedback, custom SSD1306 OLED GUI, rotary encoder and web interface (HTTP + WebSocket) • ESP-IDF / FreeRTOS
- Vision-Guided Pick-and-Place with ABB Industrial Robot (ELE610, 2025) • Computer vision (OpenCV) + ABB RAPID + Robot Web Services for autonomous pick-and-place of QR-coded objects
- LEGO Mindstorms Control Systems (ELE130, 2024) • Implemented numerical methods, FIR/IIR filters and PID controllers in MATLAB for autonomous line-following robot
- Self-Hosted Cloud • Ubuntu Server with Nextcloud / Tailscale • Secure file storage/sync

TECHNICAL SKILLS

Electronics & Embedded Systems

- ESP32, STM32, PCB design, FreeCAD, 3D printing, soldering, oscilloscope

Control & Signal Processing

- PID, state-space modeling, LQR, Kalman filtering, nonlinear control, stability, backstepping, MATLAB/Simulink, digital filters, Fourier/Laplace/Z-transform

Programming

- C/C++, Python, C#, MATLAB, JavaScript, RAPID, Structured Text

Robotics & Automation

- ABB RobotStudio, OpenCV, PLC programming, industrial communication

Other

- FreeRTOS, MQTT, Git, technical reporting

EXPERIENCE

Power Supply Engineer – UiS Aerospace (Student organization)

Summer Maintenance Worker – Grimstad Municipality (2022–2024)

Summer Assistant – Big Catering og Event (2025)

Technical Aid Service - Grimstad Municipality (2026)

LANGUAGES

Norwegian (Native) • English (Fluent) • German (Basic)